Unit 10: Nonhomogeneous Equations and Forced Response

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 10 at a Glance: Core Sub-Topics

A nonhomogeneous linear equation reads $L[y] = g(x)$, and its general solution always splits as $y = y_h + y_p$ — the complementary solution of $L[y] = 0$ plus *any* particular solution of the forcing. This unit is the toolkit for building y_p and for reading how a system answers an input.

- **Undetermined coefficients** — guess a trial of the same shape as the forcing, then match coefficients.
- **Variation of parameters** — promote the constants c_1, c_2 to functions for *any* continuous forcing.
- **The annihilator method** — an operator that kills g derives the trial form instead of memorizing it.
- **Transient and steady-state** — the homogeneous part decays, the particular part persists.
- **Unit step and impulse response** — $u(t)$, $\delta(t)$, and the system's answer to a unit kick from rest.
- **Duhamel's principle** — convolve the impulse response with the input to answer *every* forcing at once.

Part I — Condensed Cheat Sheet

10.1 The Method of Undetermined Coefficients

For a constant-coefficient $L[y] = g(x)$, the general solution is $y = y_h + y_p$: the complementary solution y_h solves $L[y] = 0$ and carries the arbitrary constants, while y_p matches the forcing. When g is an exponential, polynomial, sine/cosine, or a product of these, guess a trial of the *same shape* and solve for its coefficients.

Complementary Plus Particular

$$y = y_h + y_p, \qquad L[y_h] = 0, \qquad L[y_p] = g(x).$$

Only y_h holds the free constants, so it alone can absorb the initial or boundary conditions — which is why y_p is never the whole story.

Trial Solution Table (No Overlap)

Forcing $g(x)$	**Trial** y_p
$e^{\alpha x}$	$Ae^{\alpha x}$
polynomial of degree n	$A_n x^n + \cdots + A_1 x + A_0$
$\cos\beta x$ or $\sin\beta x$	$A\cos\beta x + B\sin\beta x$
$e^{\alpha x}\cos\beta x$	$e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

For a *sum* of forcings, build a particular solution for each piece and add them (superposition for the nonhomogeneous equation).

Common Pitfall: Overlap with the Homogeneous Solution

If any term of the trial already solves $L[y] = 0$, substitution collapses it to zero and the method fails. **Multiply the offending trial by** x (or by a higher power x^k for a root of multiplicity k) until no term overlaps y_h. For $y'' + y = \cos x$, the bare guess $A\cos x + B\sin x$ *is* the homogeneous solution; the corrected resonant trial is $x(A\cos x + B\sin x)$.

10.2 Variation of Parameters

When the forcing is not one of the friendly shapes — $\tan x$, $\sec x$, e^x/x — undetermined coefficients has nothing to guess. Variation of parameters promotes the constants c_1, c_2 of $y_h = c_1y_1 + c_2y_2$ to functions $u_1(x), u_2(x)$ and works for *any* continuous g.

Variation of Parameters Formulas

Write the equation in standard form $y'' + py' + qy = g$ (leading coefficient one), let $W = y_1y_2' - y_2y_1'$ be the Wronskian, and set $y_p = u_1y_1 + u_2y_2$ with

$$u_1' = -\frac{y_2\, g}{W}, \qquad u_2' = \frac{y_1\, g}{W},$$

$$y_p = -\, y_1 \int \frac{y_2\, g}{W}\, dx \;+\; y_2 \int \frac{y_1\, g}{W}\, dx.$$

Common Pitfall: Forgetting to Normalize

The g in these formulas is the right-hand side *after* dividing through so the y'' coefficient is one. Reading g off an unnormalized equation scales the answer incorrectly. Always normalize first, then read g.

10.3 Differential Operators and the Annihilator Method

Writing $D = \frac{d}{dx}$, a constant-coefficient operator factors like a polynomial. An **annihilator** of g is an operator $A(D)$ with $A(D)g = 0$. Applying it to both sides of $L[y] = g$ turns the problem into a larger *homogeneous* equation whose extra solutions are exactly the trial form.

Annihilator Workflow

1. Factor $L(D)$ and write the complementary solution y_h.
2. Pick the annihilator $A(D)$ of the forcing g.
3. Apply $A(D)$: solve the homogeneous equation $A(D)L(D)\,y = 0$.
4. Drop the terms already in y_h; the survivors are the trial y_p.
5. Substitute into $L[y] = g$ and match coefficients.

Common Annihilators

Function	Annihilator
$e^{\alpha x}$	$(D - \alpha)$
x^{k-1}	D^k
$\cos \beta x,\ \sin \beta x$	$(D^2 + \beta^2)$
$e^{\alpha x} \cos \beta x,\ e^{\alpha x} \sin \beta x$	$(D - \alpha)^2 + \beta^2$

A repeated factor $(D - \alpha)^k$ signals the terms $e^{\alpha x}, xe^{\alpha x}, \ldots, x^{k-1}e^{\alpha x}$. Annihilate a sum by *multiplying* the individual annihilators.

10.4 Transient and Steady-State Response

For a *stable* forced system (characteristic roots with negative real parts), the complete solution $y = y_h + y_p$ separates by long-term behavior: the homogeneous part fades while the particular part endures.

Transient Versus Steady-State

$$y(t) = \underbrace{y_h(t)}_{\text{transient } \to\ 0} + \underbrace{y_p(t)}_{\text{steady-state}}.$$

The **transient** carries the initial-condition dependence and decays because its exponentials have negative real parts. The **steady-state** is fixed by the forcing alone, so long after the start the response is essentially independent of the initial data. For sinusoidal forcing the steady state is a sinusoid at the *drive* frequency, rescaled and phase-shifted, often read off the complex gain $1/p(i\omega)$.

10.5 Unit Step and Impulse Response

Two idealized inputs model switching and striking. The unit step $u(t)$ switches a forcing on; the Dirac delta $\delta(t)$ is an instantaneous unit kick. They are linked by calculus, and so are the responses they produce.

Step, Delta, and Their Responses

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}, \qquad \delta(t) = u'(t), \qquad \int_{-\infty}^{\infty} \delta(t)\, dt = 1.$$

Sifting property: $\displaystyle\int_{-\infty}^{\infty} f(t)\,\delta(t-a)\,dt = f(a)$. The **impulse response** $w(t)$ is the output of $L[y] = \delta(t)$ from rest; the **step response** is its integral. Knowing w unlocks every input by convolution.

Common Pitfall: Forgetting the Rest State

The impulse response is defined from **rest** — zero state just before the kick. Starting with leftover energy mixes in a transient that is unrelated to the impulse, so the clean $w(t)$ (here $w(0) = 0$, $w'(0) = 1$ for a second-order system) is lost. Isolate the impulse by zeroing the initial conditions first.

10.6 Duhamel's Principle

Duhamel's principle views a general input as a continuous succession of small impulses. Each kick at time τ produces a scaled, time-shifted copy of the impulse response, and superposition sums them into a convolution.

Duhamel's Convolution

For a linear, time-invariant system starting from rest, the forced response to an input $f(t)$ is

$$y_p(t) = \int_0^t w(t-\tau)\, f(\tau)\, d\tau,$$

where w is the impulse response (the Green's function of the initial-value problem). The limits 0 to t encode **causality** — only inputs already applied can affect the present — and the argument $t - \tau$ is the time elapsed since each kick. **Linearity** and **time invariance** are exactly what make the convolution valid.

Unit 10 Key Takeaway

Every forced response is $y = y_h + y_p$. Build y_p by undetermined coefficients when the forcing is friendly (remember to multiply by x on overlap), by variation of parameters for anything continuous, or read its form from an annihilator. In a stable system the transient y_h decays and the steady-state y_p remains, and once the impulse response w is known, Duhamel's convolution answers every input at once.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through undetermined coefficients (exponential, polynomial, sinusoidal, and the two resonance traps), the annihilator method, variation of parameters for unfriendly forcing, the transient/steady-state split, and a capstone on the impulse response and Duhamel's convolution. Solutions follow in Part III.

1. **(Undetermined coefficients — exponential)** Solve $y'' - y = e^{2x}$. Give y_h, the trial, and the general solution.
2. **(Undetermined coefficients — polynomial)** Solve $y'' - y' - 2y = 4x^2$. Use a full quadratic trial.
3. **(Undetermined coefficients — sinusoid)** Solve $y'' + 4y = \sin x$. Explain why no factor of x is needed here.
4. **(Resonance, trig)** Solve $y'' + y = \cos x$. Identify the overlap with the homogeneous solution and use the corrected resonant trial.
5. **(Resonance, exponential)** Solve $y'' - 3y' + 2y = e^x$. Show why the guess gains a factor of x.
6. **(Annihilator method)** For $y'' - y' - 2y = 2e^{3x} + x$, write the annihilator of the forcing, deduce the trial form, and solve for y_p.
7. **(Variation of parameters)** Solve $y'' + y = \sec x$, where undetermined coefficients cannot be applied.
8. **(Variation of parameters, repeated root)** Find a particular solution of $y'' - 2y' + y = \dfrac{e^x}{x}$ for $x > 0$.
9. **(Transient and steady-state IVP)** Solve $y'' + 4y' + 3y = 6$, $y(0) = 0$, $y'(0) = 0$. Identify the transient and the steady-state pieces.
10. **(Impulse response and Duhamel, capstone)** Find the impulse response $w(t)$ of $y'' + 3y' + 2y = f(t)$ (rest initial conditions), then write Duhamel's convolution for a general input f.

Part III — Complete Solutions

Solution 1. The homogeneous equation $y'' - y = 0$ has roots $r = \pm 1$, so $y_h = c_1 e^x + c_2 e^{-x}$. Since e^{2x} does not overlap y_h, try $y_p = Ae^{2x}$, so $y_p'' = 4Ae^{2x}$:

$$4Ae^{2x} - Ae^{2x} = 3Ae^{2x} = e^{2x} \;\Rightarrow\; A = \tfrac{1}{3}.$$

$$y = c_1 e^x + c_2 e^{-x} + \tfrac{1}{3} e^{2x}.$$

Solution 2. Roots of $r^2 - r - 2 = (r-2)(r+1) = 0$ give $y_h = c_1 e^{2x} + c_2 e^{-x}$. Use the complete quadratic trial $y_p = Ax^2 + Bx + C$, with $y_p' = 2Ax + B$, $y_p'' = 2A$:

$$2A - (2Ax + B) - 2(Ax^2 + Bx + C) = -2Ax^2 + (-2A - 2B)x + (2A - B - 2C).$$

Matching $-2Ax^2 + \cdots = 4x^2$ term by term:

$$-2A = 4 \Rightarrow A = -2; \quad -2A - 2B = 0 \Rightarrow B = 2; \quad 2A - B - 2C = 0 \Rightarrow C = -3.$$

$$y = c_1 e^{2x} + c_2 e^{-x} - 2x^2 + 2x - 3.$$

Solution 3. Here $y'' + 4y = 0$ has roots $\pm 2i$, so $y_h = c_1 \cos 2x + c_2 \sin 2x$. The forcing $\sin x$ has frequency $1 \neq 2$, so there is *no* overlap and no factor of x is needed. Try $y_p = A\cos x + B\sin x$; then $y_p'' = -A\cos x - B\sin x$:

$$(-A + 4A)\cos x + (-B + 4B)\sin x = 3A\cos x + 3B\sin x = \sin x.$$

So $A = 0$, $B = \frac{1}{3}$, giving

$$y = c_1 \cos 2x + c_2 \sin 2x + \tfrac{1}{3}\sin x.$$

Solution 4. For $y'' + y = 0$ the roots are $\pm i$, so $y_h = c_1 \cos x + c_2 \sin x$. The natural guess $A\cos x + B\sin x$ *is* y_h — a resonance. Multiply by x: $y_p = x(A\cos x + B\sin x)$. Differentiating, $y_p'' = -x(A\cos x + B\sin x) + 2(-A\sin x + B\cos x)$, so

$$y_p'' + y_p = 2(-A\sin x + B\cos x) = \cos x \;\Rightarrow\; B = \tfrac{1}{2},\; A = 0.$$

$$y = c_1 \cos x + c_2 \sin x + \tfrac{x}{2}\sin x.$$

Solution 5. Roots of $r^2 - 3r + 2 = (r-1)(r-2) = 0$ give $y_h = c_1 e^x + c_2 e^{2x}$. The forcing e^x overlaps the e^x solution, so the bare guess Ae^x fails; multiply by x: $y_p = Axe^x$. Then $y_p' = Ae^x(1+x)$ and $y_p'' = Ae^x(2+x)$, so

$$Ae^x\big[(2+x) - 3(1+x) + 2x\big] = Ae^x(-1) = e^x \;\Rightarrow\; A = -1.$$

$$y = c_1 e^x + c_2 e^{2x} - xe^x.$$

Solution 6. The operator is $L(D) = D^2 - D - 2 = (D-2)(D+1)$, so $y_h = c_1 e^{2x} + c_2 e^{-x}$. The forcing $2e^{3x} + x$ is annihilated by

$$A(D) = (D-3)\,D^2,$$

since $(D-3)$ kills e^{3x} and D^2 kills the linear term x. Neither e^{3x} nor $\{1, x\}$ overlaps y_h, so the trial is $y_p = Ae^{3x} + Bx + C$. Substitute the exponential piece Ae^{3x} ($y'' - y' - 2y = 9A - 3A - 2A = 4A$):

$$4Ae^{3x} = 2e^{3x} \;\Rightarrow\; A = \tfrac{1}{2}.$$

For the polynomial piece $Bx + C$ ($y'' = 0$, $y' = B$):

$$-B - 2(Bx + C) = -2Bx + (-B - 2C) = x \;\Rightarrow\; B = -\tfrac{1}{2},\; C = \tfrac{1}{4}.$$

$$y = c_1 e^{2x} + c_2 e^{-x} + \tfrac{1}{2}e^{3x} - \tfrac{1}{2}x + \tfrac{1}{4}.$$

Solution 7. The equation is already in standard form with $y_1 = \cos x$, $y_2 = \sin x$, and Wronskian $W = \cos x \cos x - \sin x(-\sin x) = 1$. With $g = \sec x$,

$$u_1' = -\frac{y_2 g}{W} = -\sin x \sec x = -\tan x \;\Rightarrow\; u_1 = \ln|\cos x|,$$

$$u_2' = \frac{y_1 g}{W} = \cos x \sec x = 1 \;\Rightarrow\; u_2 = x.$$

Hence $y_p = u_1 y_1 + u_2 y_2$:

$$y = c_1 \cos x + c_2 \sin x + \cos x\, \ln|\cos x| + x \sin x.$$

Solution 8. The repeated root $r = 1$ of $(r-1)^2$ gives $y_1 = e^x$, $y_2 = xe^x$, with $W = e^x(e^x + xe^x) - xe^x \cdot e^x = e^{2x}$. In standard form $g = e^x/x$, so

$$u_1' = -\frac{xe^x \cdot (e^x/x)}{e^{2x}} = -1 \Rightarrow u_1 = -x, \qquad u_2' = \frac{e^x \cdot (e^x/x)}{e^{2x}} = \frac{1}{x} \Rightarrow u_2 = \ln x.$$

Then $y_p = -xe^x + xe^x \ln x$; the term $-xe^x$ already lives in y_h, so

$$y = (c_1 + c_2 x)e^x + xe^x \ln x.$$

Solution 9. Roots of $r^2 + 4r + 3 = (r+1)(r+3) = 0$ are $r = -1, -3$, both negative, so $y_h = c_1 e^{-t} + c_2 e^{-3t}$ is the **transient**. A constant forcing calls for $y_p = K$ with $3K = 6$, so the **steady-state** is $y_p = 2$. Apply $y(0) = 0$, $y'(0) = 0$ to $y = c_1 e^{-t} + c_2 e^{-3t} + 2$:

$$c_1 + c_2 + 2 = 0, \qquad -c_1 - 3c_2 = 0 \;\Rightarrow\; c_1 = -3c_2.$$

$$-3c_2 + c_2 + 2 = 0 \Rightarrow c_2 = 1,\; c_1 = -3, \qquad y = -3e^{-t} + e^{-3t} + 2.$$

As $t \to \infty$ the transient $-3e^{-t} + e^{-3t} \to 0$, leaving the steady-state value 2.

Solution (Capstone). The impulse response solves $w'' + 3w' + 2w = \delta(t)$ from rest, which for a second-order system means $w(0) = 0$, $w'(0) = 1$. Roots of $r^2 + 3r + 2 = (r+1)(r+2)$ are $r = -1, -2$, so $w = Ae^{-t} + Be^{-2t}$:

$$w(0) = A + B = 0, \qquad w'(0) = -A - 2B = 1 \;\Rightarrow\; B = -1,\; A = 1.$$

$$w(t) = e^{-t} - e^{-2t}, \quad t \geq 0.$$

By Duhamel's principle the forced response from rest to any input f is

$$y_p(t) = \int_0^t w(t-\tau)\, f(\tau)\, d\tau = \int_0^t \left(e^{-(t-\tau)} - e^{-2(t-\tau)}\right) f(\tau)\, d\tau.$$

Unit 10 Key Takeaway

The shape of the forcing chooses the tool: undetermined coefficients for exponentials, polynomials, and sinusoids — with the $\times x$ resonance fix whenever a trial term solves $L[y] = 0$ — variation of parameters for $\sec x$, e^x/x, and other unfriendly inputs, and the annihilator to derive the trial form. In a stable system the transient decays to the steady-state, and the impulse response $w(t) = e^{-t} - e^{-2t}$ feeds Duhamel's convolution to resolve every input at once.